

SULIT



SECP2523 DATABASE

SESSION 2024/2025 - SEMESTER 1

ALTERNATIVE ASSESSMENT REPORT: PHASE 1

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Case Study	WBL Project
System Name	KADA COOPERATIVE SYSTEM
Group Name	KODELAB

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Section A: System's Overview

1.1 Overall Description

The system that we have proposed in the WBL project is KADA Cooperative System in which we aim to streamline the cooperative's current manual process, especially focusing on the membership and loan application by transitioning it to the new web-based platform. Module that I was assigned by my groupmate in this WBL project is module authentication and module membership.

Firstly, in the authentication module, we have implemented a user login and reset password function to this module. The user can login to the system by choosing their roles in the cooperative's and key in their identity card number and password to the system for authentication process. Since, we implemented a multi-role authentication, it will redirect the user to their role main page accordingly. Else, if the user is invalid after they have tried to login to the system, the system will deny an access from that unauthorized user. Thus, it is crucial to have an authentication module in this system as it will ensure only the authorized user can access the system by verifying their credentials and to protect the user personal information such as membership and loan information. Also, for the authentication module, we implemented the reset password function as a way for the user to recover their password by key in their email and new password to the system. The user needs to provide their registered email for credential or the system will refuse the user to change their password. This is to make sure that the user has been given the access to use the system to prevent any unauthorized individuals from accessing the system.

Next, for the membership module, we have implemented an online form which is transforming from the hand written form of membership application. It is easier and more convenient for the user to apply for membership online and for the cooperative's to keep their applicant's information in their database. The user will get an email showing that their application status is accepted or rejected by the board of director or admin after the cooperative's internal meeting. This module helps to reduce the paper based application and easier for the cooperative's board of directors to review all membership applications at one click. The user needs to register for this module first before they get to login to the system as the password will be given through their email after their application status is accepted. This module is the modernized module for the membership application and helps the board of directors to check the validity of their information.

Lastly, table 1.1.1 shows the module distribution of the WBL project for my group which is KodeLab.

Table 1.1.1 Module Distribution

Group Members' Name	Matric No	Module
Nabil Aflah Boo Binti Mohd Yosuf Boo Yong Chong	A23CS0252	Authentication
		Membership
Lubna Al Haani Binti Radzuan	A23CS0107	Loan
		Approval
Nurul Adriana Binti Kamal Jefri	A23CS0258	Reporting
Nur Firzana Binti Badrus Hisham	A23CS0156	Admin
Pravinraj A/L Sivabathi	A23CS0171	Statement

1.2 Project Weaknesses or Improvement

Some of the weaknesses that can be seen from the authentication module is that the role based login in which the user is required to choose their role in the login page asides enter their ic number and password. To have a better user experience to login the system, the role selection part should be removed and the user can just login according to their ic number and password. This is because the role selection part seems to be an additional step for the user to login the system as the system should identify the user role based on their ic number and password as in the cooperative database. Therefore, the authentication module should be modified for the user to login by ic number and password that they have registered and received upon the membership application. Another improvement that can be made in this module is to implement an OTP for the password reset after the user enters their email address. It enhances the security and data protection through a two-step authentication process. In addition, one of the weaknesses in the membership module is that there is a lack of document uploading for the membership application. By adding a document attachment feature in the membership application, it helps to validate the trustworthiness of the salary information. Therefore, it ensures the integrity of the membership information. Another improvement can be made for this module is by sending an email notification containing a link to track the application progress so that the user won't feel left out after they made the registration.

1.3 Proposed Module

The new modules that I propose to overcome the weaknesses are OTP module and tracking membership application module. For the OTP module, it helps to enhance the security of the password reset function even if their email address is compromised. Flow of this module is the system will first validate the email address for the password reset and if the email exists in the system, it will generate a one time password (OTP) and send via the user email. The OTP is valid for the limited time for the user to verify and the system will prompt the user to reset their password if the OTP entered is valid. Key feature of this module is to limit the user to enter the OTP for maximum 3 attempts or else the system will lock the user account.

Next, the document management module allows the KADA staff to upload required documents such as identity card or income proof through the system which make it easier for the admin and board of directors to review the validity of the staff information. The flow of the module is that the membership application form will require the user to upload the document needed as a proof and submit it to the system. The admin then will review the submitted document and will approve the document if it meets the requirement or else the admin will send a comment to the applicant via email for them to reupload the document as the document is blurry. This module features a real-time file validation and detail tracking of approval of the membership application.

Section B: Database Planning and Design

2.1 Conceptual ERD

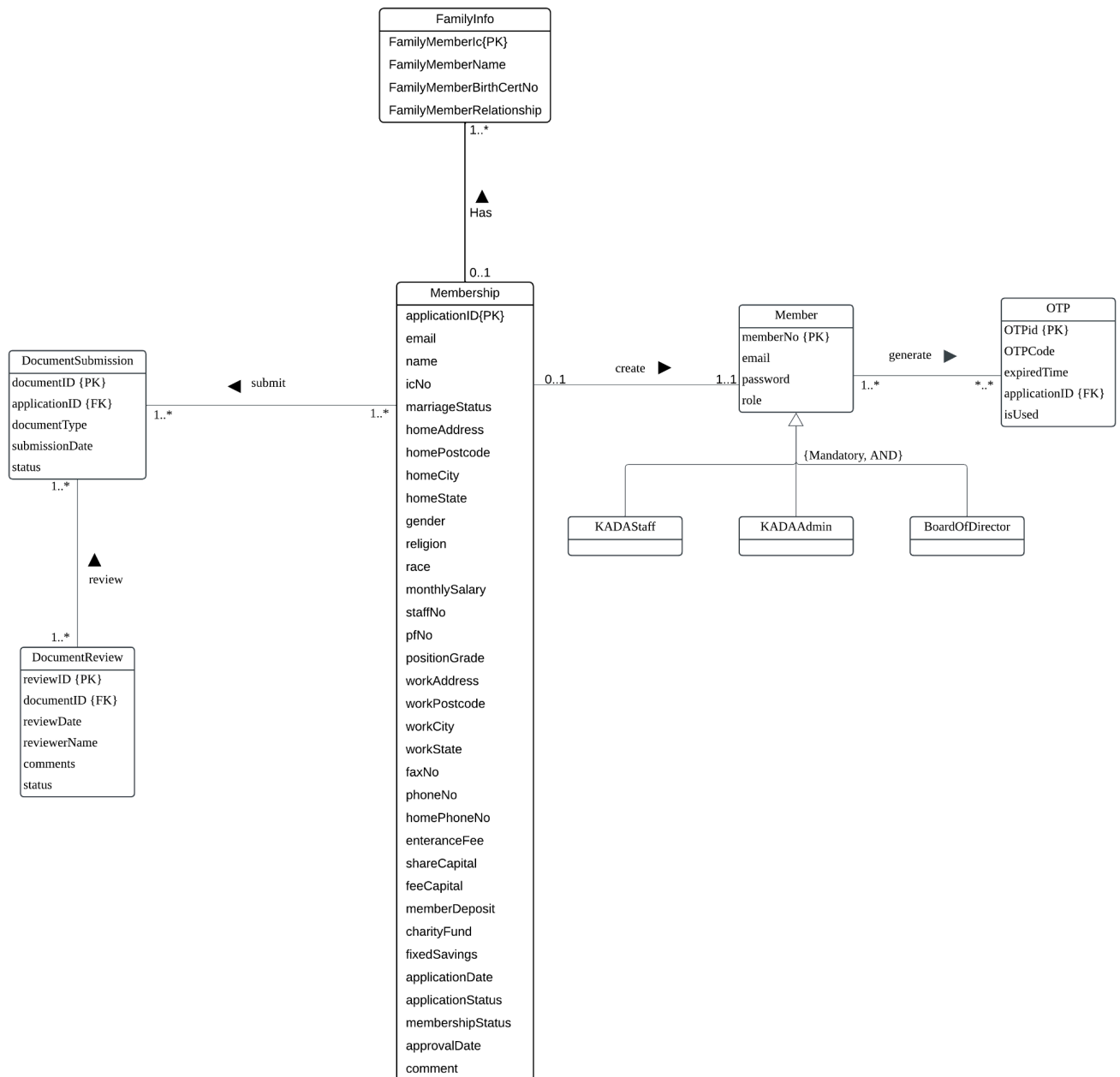


Figure 2.1.1 Conceptual ERD for Module OTP and Module Document Management

2.2 Logical ERD

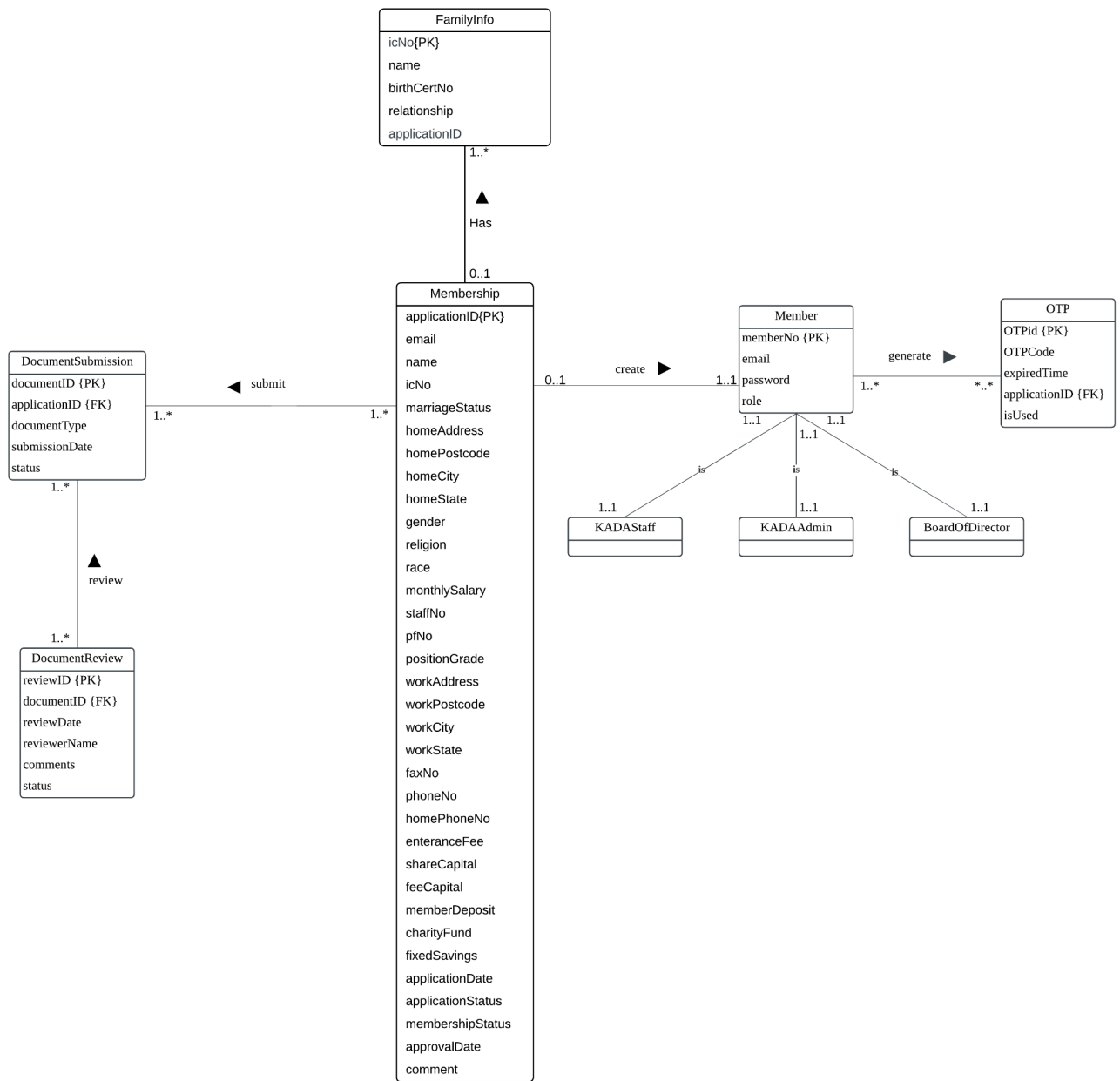


Figure 2.2.1 Logical ERD of Module OTP and Module Document Management

2.3 Data Dictionary For Logical ERD

Member table

Attribute	Description	Data Type	Length	Constraints
memberNo	Unique number of each staff member	INT	10	Primary Key, NOT NULL
email	Email address corresponding with the user	VARCHAR	255	Unique, NOT NULL
password	Encrypted password for user authentication	VARCHAR	8	NOT NULL
role	User role in the cooperative	VARCHAR	10	NOT NULL

OTP table

Attribute	Description	Data Type	Length	Constraints
OTPid	Unique identifier for each OTP	INT	10	Primary Key, NOT NULL
OTPCode	One Time Password code for each user to reset password	INT	6	NOT NULL
expiredTime	Timestamp of the validation OTP code	DATETIME	-	NOT NULL
applicationID	Unique identifier for each user	INT	10	Foreign Key, NOT NULL
isUsed	Indicates if the code has been used	BOOLEAN	-	NOT NULL

Membership table

Attribute	Description	Data Type	Length	Constraints
applicationID	Unique identifier for each user	INT	10	Primary Key, NOT NULL
email	Email address correspond to each user	VARCHAR	255	NOT NULL
name	Full name of each user	VARCHAR	255	NOT NULL
icNo	Identification card number of each user	INT	15	Unique, NOT NULL
marriageStatus	Marriage status of each user	VARCHAR	255	NOT NULL
homeAddress	Home address of each user	VARCHAR	255	NOT NULL
homePostcode	Postcode of the user home address	INT	5	NOT NULL
homeCity	City of the user home address	VARCHAR	255	NOT NULL
homeState	State of the user home address	VARCHAR	255	NOT NULL
gender	User gender	VARCHAR	10	NOT NULL
religion	User religion	VARCHAR	20	NOT NULL
race	User race	VARCHAR	20	NOT NULL
monthlySalary	User salary of each month	DECIMAL	12, 2	NOT NULL
staffNo	Staff number of the user in the cooperative	INT	5	Unique, NOT NULL
pfNo	Pf number of each staff in the cooperative	INT	5	NOT NULL

positionGrade	Position and grade of each staff in the cooperative	VARCHAR	255	NOT NULL
workAddress	Work address of the user	VARCHAR	255	NOT NULL
workPostcode	Postcode of the work address	INT	5	NOT NULL
workCity	City of the work address	VARCHAR	255	NOT NULL
workState	State of the work address	VARCHAR	255	NOT NULL
faxNo	Fax number of the user	INT	10	NOT NULL
phoneNo	Phone number of the user	INT	12	NOT NULL
homephoneNo	Home phone number of the user	INT	12	NOT NULL
entranceFee	Entrance fee that user wish to give for minimum RM35	DECIMAL	12, 2	NOT NULL
shareCapital	Share capital fee that user willing to give	DECIMAL	12, 2	NOT NULL
feeCapital	Fee capital that the user willing to give	DECIMAL	12, 2	NOT NULL
memberDeposit	Member deposit fee that the user want to give	DECIMAL	12, 2	NOT NULL
charityFund	Charity fund that the user wish to give	DECIMAL	12, 2	NOT NULL
fixedSavings	Fixed savings of the user	DECIMAL	12, 2	NOT NULL
applicationDate	Submission date of the membership application	DATE	-	NOT NULL

applicationStatus	Status of the membership application	VARCHAR	20	NOT NULL
approvalDate	Approval date of the membership application	DATE	-	NOT NULL
comment	Comments from the admin or board of directors of the membership application	TEXT	-	NOT NULL

FamilyInfo table

Attribute	Description	Data Type	Length	Constraints
icNo	Identity card number of the family member	INT	15	Primary Key, NOT NULL
name	Full name of the family member	VARCHAR	255	NOT NULL
birthCertNo	Birth certificate number of the family member	VARCHAR	8	NULL
relationship	Relationship of the family member and the user	VARCHAR	255	NOT NULL
applicationID	Foreign key link to membership	INT	10	Foreign Key, NOT NULL

DocumentSubmission table

Attribute	Description	Data Type	Length	Constraints
documentID	Unique identifier of each document	INT	10	Primary Key, NOT NULL
applicationID	Foreign key link to the membership	INT	10	Foreign Key, NOT NULL

	application			
documentType	Type of the document	VARCHAR	255	NOT NULL
submissionDate	Timestamp of the submitted document	DATE	-	NOT NULL
status	Status of the document	VARCHAR	255	NOT NULL

DocumentReview Table

Attribute	Description	Data Type	Length	Constraints
reviewID	Unique identifier of each document review	INT	10	Primary Key, NOT NULL
documentID	Foreign key link to the submitted document	INT	10	Foreign Key, NOT NULL
reviewDate	Timestamp of the document reviewed	DATE	-	NOT NULL
reviewerName	Name of the reviewer	VARCHAR	255	NOT NULL
comments	Feedback provided during the review	TEXT	-	NOT NULL
status	Review status	VARCHAR	20	NOT NULL

2.4 Relational Database Schema

Membership(applicationID, email, name, icNo, marriageStatus, homeAddress, homePostcode, homeCity, homeState, gender, religion, race, monthlySalary, staffNo, pfNo, positionGrade, workAddress, workPostcode, workCity, workState, faxNo, phoneNo, homePhoneNO, entranceFee, shareCapital, feeCapital, memberDeposit, charityFund, fixedSavings, applicationDate, applicationStatus, membersipStatus, approvalDate, comment)

Primary key: applicationID

Member(memberNo, icNo, email, password, role, applicationID)

Primary key: memberNo

Foreign key: memberNo REFERENCES Membership(staffNo)

Foreign key: icNo REFERENCES Membership(icNo)

Foreign key: applicationID REFERENCES Membership(applicationID)

DocumentSubmission(documentID, documentType, submissionDate, status, applicationID)

Primary key: documentID

Foreign key: applicationID REFERENCES Membership(applicationID)

DocumentReview(reviewID, reviewDate, reviewName, comments, status, documentID)

Primary key: reviewID

Foreign key: documentID REFERENCES DocumentSubmission(documentID)

FamilyInfo(icNo, name, birthCertNo, relationship, applicationID)

Primary key: icNo

Foreign key: applicationID REFERENCES Membership(applicationID)

OTP(OTPid, OTPCode, expiredTime, isUsed, applicationID)

Primary key: OTPid

Foreign key: applicationID REFERENCES Membership(applicationID)

2.5 Normalization

1. 1NF:

Membership(applicationID, email, name, icNo, marriageStatus, homeAddress, homePostcode, homeCity, homeState, gender, religion, race, monthlySalary, staffNo, pfNo, positionGrade, workAddress, workPostcode, workCity, workState, faxNo, phoneNo, homePhoneNO, entranceFee, shareCapital, feeCapital, memberDeposit, charityFund, fixedSavings, applicationDate, applicationStatus, membersipStatus, approvalDate, comment)

Primary key: applicationID

2NF: No partial dependencies.

3NF: No transitive dependencies.

Thus, up until 3NF,

Membership(applicationID, email, name, icNo, marriageStatus, homeAddress, homePostcode, homeCity, homeState, gender, religion, race, monthlySalary, staffNo, pfNo, positionGrade, workAddress, workPostcode, workCity, workState, faxNo, phoneNo, homePhoneNO, entranceFee, shareCapital, feeCapital, memberDeposit, charityFund, fixedSavings, applicationDate, applicationStatus, membersipStatus, approvalDate, comment)

Primary key: applicationID

2. 1NF:

Member(memberNo, icNo, email, password, role, applicationID)

Primary key: memberNo

Foreign key: memberNo REFERENCES Membership(staffNo)

Foreign key: icNo REFERENCES Membership(icNo)

Foreign key: applicationID REFERENCES Membership(applicationID)

2NF: No partial dependencies.

3NF: No transitive dependencies.

Thus, up until 3NF,

Member(memberNo, icNo, email, password, role)

Primary key: memberNo

3. 1NF:

DocumentSubmission(documentID, documentType, submissionDate, status, applicationID)

Primary key: documentID

Foreign key: applicationID REFERENCES Membership(applicationID)

2NF: No partial dependencies.

3NF: No transitive dependencies.

Thus, up until 3NF,

DocumentSubmission(documentID, documentType, submissionDate, status, applicationID)

Primary key: documentID

Foreign key: applicationID REFERENCES Membership(applicationID)

4. 1NF:

DocumentReview(reviewID, reviewDate, reviewName, comments, status, documentID)

Primary key: reviewID

Foreign key: documentID REFERENCES DocumentSubmission(documentID)

2NF: No partial dependencies

3NF: No transitive dependencies

Thus, up until 3NF,

DocumentReview(reviewID, reviewDate, reviewName, comments, status, documentID)

Primary key: reviewID

Foreign key: documentID REFERENCES DocumentSubmission(documentID)

5. 1NF:

FamilyInfo(icNo, name, birthCertNo, relationship, applicationID)

Primary key: icNo

Foreign key: applicationID REFERENCES Membership(applicationID)

2NF: No partial dependencies

3NF: No transitive dependencies

Thus, up until 3NF,

FamilyInfo(icNo, name, birthCertNo, relationship, applicationID)

Primary key: icNo

Foreign key: applicationID REFERENCES Membership(applicationID)

6. 1NF:

OTP(OTPid, OTPCode, expiredTime, isUsed, applicationID)

Primary key: OTPid

Foreign key: applicationID REFERENCES Membership(applicationID)

2NF: No partial dependencies

3NF: No transitive dependencies

Thus, up until 3NF,

OTP(OTPid, OTPCode, expiredTime, isUsed, applicationID)

Primary key: OTPid

Foreign key: applicationID REFERENCES Membership(applicationID)